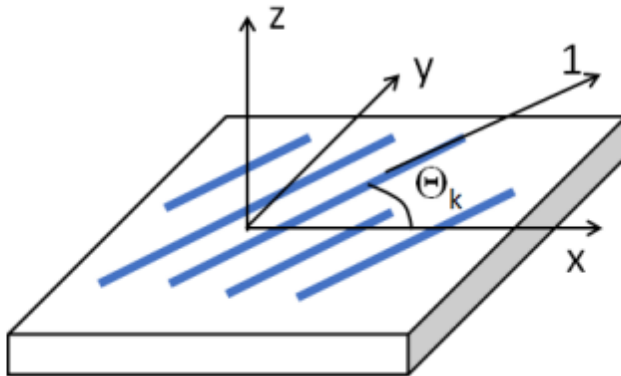
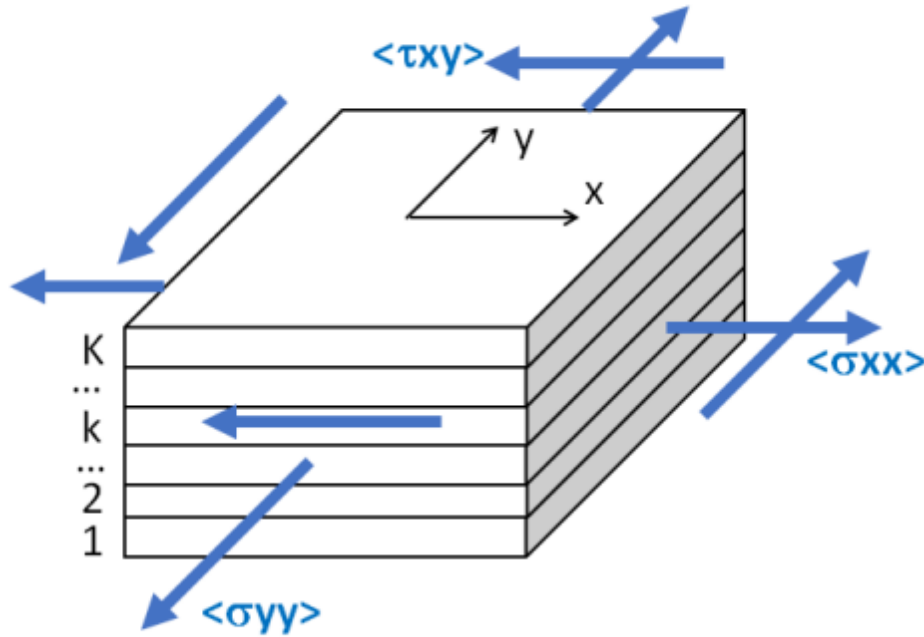


Laminated Strength Analysis

$[+\Theta/-\Theta/+30^\circ/-30^\circ]_s$ SM Carbon Composite | Hashin (1980) Failure Criterion

Problem Statement



- **Objective**
Analyze first-ply failure (FPF) of a symmetric laminated composite under combined in-plane loading using the Hashin (1980) failure criterion.
- **Laminate Configuration**
Symmetric layup: $[+\Theta/-\Theta/+\alpha/-\alpha]_s$
 $\Theta = 67.8^\circ$, $\alpha = 30^\circ$
8 plies, each $h = 0.5$ mm $\rightarrow$ total thickness $H = 4$ mm
- **Applied In-Plane Stresses**
 $\langle \sigma_{xx} \rangle = 2099$ MPa
 $\langle \sigma_{yy} \rangle = 620$ MPa
 $\langle \tau_{xy} \rangle = 507$ MPa

Input Data

Layup: $[+67.8^\circ / -67.8^\circ / +30^\circ / -30^\circ]_s$

8 plies

$h_{\text{ply}} = 0.5 \text{ mm}$

$H_{\text{total}} = 4.0 \text{ mm}$

Elastic Properties (SM Carbon)

Property	Value
Axial modulus, E_{11}	145,000 MPa
Transverse modulus, E_{22}	10,000 MPa
Shear modulus, G_{12}	4,100 MPa
Major Poisson's ratio, ν_{12}	0.25
Minor Poisson's ratio, ν_{21}	0.01724

Strength Properties (SM Carbon)

Property	Value
Axial tension, R_{11}^T	1520 MPa
Axial compression, R_{11}^C	1380 MPa
Transverse tension, R_{22}^T	41 MPa
Transverse compression, R_{22}^C	170 MPa
In-plane shear, T_{12}	80 MPa

Algorithm of Analysis (Hashin Criterion)

Step 1 – Compute reduced stiffness matrix $\mathbf{Q}$ from E_{11} , E_{22} , ν_{12} , G_{12} in material coordinates (1-2).

Step 2 – Apply transformation $\mathbf{T}(\Theta_k)$ to obtain transformed stiffness $\mathbf{Q}_k$ for each ply orientation Θ_k in global x-y frame.

Step 3 – Assemble extensional stiffness matrix $[\mathbf{A}] = \sum \mathbf{Q}_k \cdot h_k$ for all 8 plies.

Step 4 – Compute mid-plane strains $\{\epsilon^0\} = [\mathbf{A}]^{-1} \cdot \{\mathbf{N}\}$, where force resultants $\{\mathbf{N}\} = \{\langle o \rangle\} \cdot H$.

Step 5 – Back-calculate ply stresses in global frame $\{\sigma\}_{\text{global}} = \mathbf{Q}_k \cdot \{\epsilon^0\}$, then rotate to material coordinates $\{\sigma\}_{\text{material}} = \mathbf{T}(\Theta_k) \cdot \{\sigma\}_{\text{global}}$.

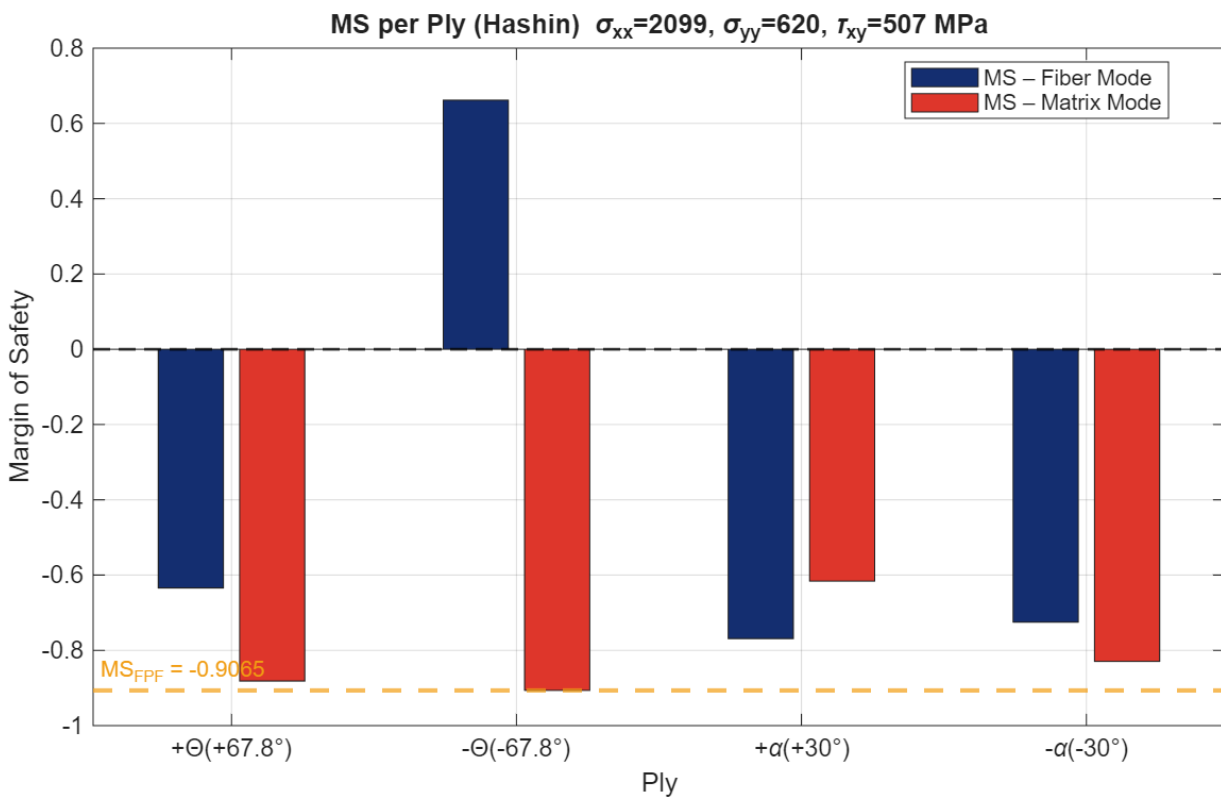
Step 6 – Apply Hashin criterion to compute Failure Index (FI) for fiber and matrix modes.

Margin of Safety: $MS = 1/\sqrt{FI} - 1$.

First Ply Failure MS_0 = minimum MS over all plies.

Hashin Failure Indices		
Mode	Condition	FI
Fiber tension	$\sigma_1 \geq 0$	$(\sigma_1/R_{11}^t)^2 + (\tau_{12}/T_{12})^2$
Fiber compression	$\sigma_1 < 0$	$(\sigma_1/R_{11}^c)^2$
Matrix tension	$\sigma_2 \geq 0$	$(\sigma_2/R_{22}^t)^2 + (\tau_{12}/T_{12})^2$
Matrix compression	$\sigma_2 < 0$	$(\sigma_2/2T_{12})^2 + ((R_{22}^c/2T_{12})^2 - 1) + (\tau_{12}/T_{12})^2$

Ply	σ_1 (MPa)	σ_2 (MPa)	τ_{12} (MPa)	MS Fiber	Mode	MS Matrix	Mode	MS Min
+ Θ (+67.8°)	1208.35	328.78	-209.30	-0.6343	FT	-0.8814	MT	-0.8814
- Θ (-67.8°)	-830.45	436.09	90.01	+0.6617	FC	-0.9065	MT	-0.9065
+ α (+30°)	5984.60	77.40	-143.50	-0.7689	FT	-0.6160	MT	-0.7689
- α (-30°)	3461.02	210.22	226.98	-0.7251	FT	-0.8294	MT	-0.8294



Ply Stresses & Margins of Safety

$\langle \sigma_{xx} \rangle = 2099$ Mpa

$\langle \sigma_{yy} \rangle = 620$ Mpa

$\langle \tau_{xy} \rangle = 507$ Mpa

Note on symmetry: The layup is symmetric. Plies 5–8 repeat plies 4–1 in reverse order: Ply5 = Ply4, Ply6 = Ply3, Ply7 = Ply2, Ply8 = Ply1. Only the four unique plies are shown above.

Observations:

- The **- Θ** ply shows a **positive MS Fiber (+0.6617)** because it is in compression ($\sigma_1 < 0$) and the compressive strength is high.
- However, the **matrix mode** for this same ply is strongly negative (**-0.9065**), which governs the **First Ply Failure (FPF)**.
- FPF $MS_0 = -0.9065$** (driven by matrix tension in the -67.8° ply).

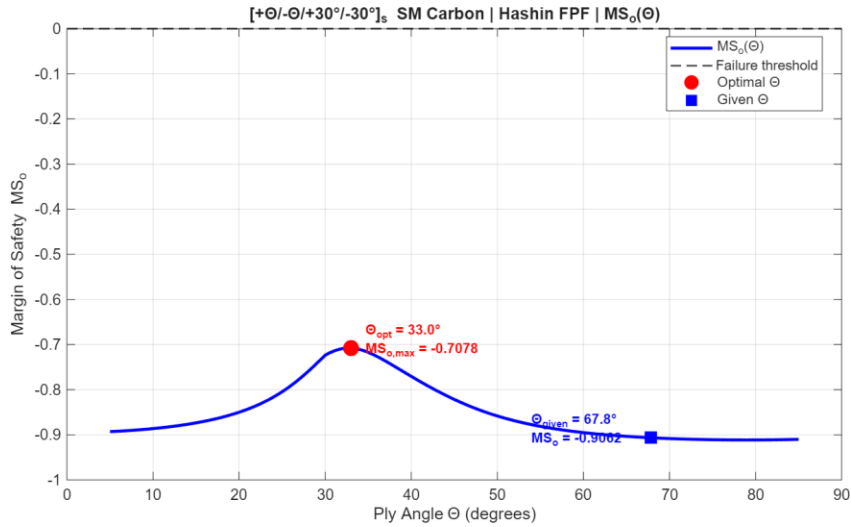
Laminated Strength Prediction

Laminate Strength (First Ply Failure, unit-stress method)	
Strength	Value (MPa)
⟨Rxx⟩ (axial tensile strength)	213.7 MPa
⟨Ryy⟩ (transverse tensile strength)	328.6 MPa
⟨Txy⟩ (in-plane shear strength)	297.9 MPa

Laminate Strength (First Ply Failure, unit-stress method)

All three load components individually exceed the laminate’s FPF-based strength

Comparison with applied loads				
Direction	Laminate Strength (MPa)	Applied Load (MPa)	Ratio (Applied/Strength)	Status
x (axial)	213.7	2099	9.82	FAILED
y (transverse)	328.6	620	1.89	FAILED
shear	297.9	507	1.70	FAILED



Optimization results ($\alpha = 30^\circ$ fixed)

Parameter	Value
Given Θ	67.8°
MS_0 at given Θ	-0.9062
Optimal Θ	33.0°
$MS_{0,max}$ at optimal Θ	-0.7078
Improvement	0.1984 ($\approx 22\%$ relative)

MS_0 vs. Ply Angle Θ

The laminate fails for all Θ under the given load magnitude, but the optimal angle provides a 22% higher margin

Calculation Tool (MATLAB R2024a)

Code structure

- Sections 1-2: Material properties and reduced stiffness Q
- Section 3: $[A]$ matrix assembly and inversion
- Section 4: Mid-plane strains from force resultants
- Section 5: Ply-level stress transformation and Hashin MS calculation
- Section 6: Laminate strength via unit-stress method
- Sections 7-10: Θ -sweep, optimization, and figure generation

Local functions

fn_Qbar – transformed stiffness
fn_Tstress – stress transformation
fn_hashin – Hashin failure indices and MS
fn_MS_FPF – FPF margin for arbitrary load
fn_bisect_syy – binary search for failure envelope

Output files

- chart_ms_theta_matlab.png
- chart_msBars_matlab.png
- chart_failure_envelopes_matlab.png

```

62 % 4. MID-PLANE STRAINS
63 %%
64 N_res = [Sxx_app; Syy_app; Sxy_app] * H_lam; % force resultants (N/mm)
65 eps0 = A_inv * N_res;
66
67 fprintf('==== MID-PLANE STRAINS =====\n');
68 fprintf('  eps_xx = %.6f\n  eps_yy = %.6f\n  gam_xy = %.6f\n\n', ...
69     eps0(1), eps0(2), eps0(3));
70
71 %%
72 % 5. PLY-LEVEL STRESSES & HASHIN MARGINS
73 %%
74 ply_lbl = {'+\Theta(+67.8^\circ)', '-\Theta(-67.8^\circ)', '+\alpha(+30^\circ)', '-\alpha(-30^\circ)', ...
75     '-\alpha(-30^\circ)', '+\alpha(+30^\circ)', '-\Theta(-67.8^\circ)', '+\Theta(+67.8^\circ)'};
76
77 fprintf('==== PLY STRESSES AND MARGINS OF SAFETY =====\n');
78 hdr = '%-20s %9s %9s | %9s %-4s | %9s %-4s | %9s';
79 fprintf([hdr ' \n'], 'Ply', 'sig1', 'sig2', 'tau12', 'MS_fib', 'mode', 'MS_mat', 'mode', 'MS_min');
80 fprintf('%s\n', repmat('-', 1, 88));
81
82 MSf_all = zeros(1, nPly);
83 MSm_all = zeros(1, nPly);
84 MSmin_all = zeros(1, nPly);
85 mf_all = cell(1, nPly);
86 mm_all = cell(1, nPly);
87
88 for k = 1:nPly
89     sg = Qb{k} * eps0;
90     T = fn_Tstress(pang(k));
91     sm = T * sg;
92     s1 = sm(1); s2 = sm(2); t12 = sm(3);
93
94     [MSf, MSm, mf, mm] = fn_hashin(s1, s2, t12, F1T, F1C, F2T, F2C, F12);
95     MSmin = min(MSf, MSm);
96
97     MSf_all(k) = MSf;
98     MSm_all(k) = MSm;
99     MSmin_all(k) = MSmin;
100    mf_all(k) = mf;
101    mm_all(k) = mm;
102
103    fprintf('%-20s %9.2f %9.2f | %9.4f %-4s | %9.4f %-4s | %9.4f\n', ...
104        ply_lbl{k}, s1, s2, t12, MSf, mf, MSm, mm, MSmin);
105 end
106
107 MS_FPF = min(MSmin_all);
108 fprintf('%s\n', repmat('-', 1, 88));
109 fprintf('%-20s %s %9.4f\n', 'FPF', 'MS_o =', MS_FPF);
110
111 %%
112 % 6. LAMINATED STRENGTH (<Rxx>, <Ryy>, <Txy>)
113 %%
114 MS_u_xx = fn_MS_FPF(1, 0, 0, pang, Qb, A_inv, H_lam, F1T, F1C, F2T, F2C, F12);
115 MS_u_yy = fn_MS_FPF(0, 1, 0, pang, Qb, A_inv, H_lam, F1T, F1C, F2T, F2C, F12);
116 MS_u_xy = fn_MS_FPF(0, 0, 1, pang, Qb, A_inv, H_lam, F1T, F1C, F2T, F2C, F12);
117
118 Rxx_lam = (MS_u_xx + 1) * 1.0;
119 Ryy_lam = (MS_u_yy + 1) * 1.0;
120 Txy_lam = (MS_u_xy + 1) * 1.0;
121
122 fprintf('\n===== LAMINATED STRENGTH =====\n');

```

```

1 %% =====
2 % HW4 - Laminated Strength | Aerospace Materials | Spring 2026
3 % Student : Krishiv A. Patel
4 % Material : SM Carbon | Layup: [+67.8/-67.8/+30/-30]s
5 % Loading : <sxx>=2099 MPa, <syy>=620 MPa, <txy>=507 MPa
6 % Criterion: Hashin (1980)
7 % =====
8 clc; clear; close all;
9
10 %%
11 % 1. MATERIAL & GEOMETRIC INPUT
12 %%
13 % Elastic constants (MPa)
14 E1 = 145e3; E2 = 10e3;
15 G12 = 4.1e3; nu12 = 0.25;
16 nu21 = nu12 * E2 / E1;
17
18 % Strength parameters (MPa)
19 F1T = 1520; F1C = 1380;
20 F2T = 41; F2C = 170;
21 F12 = 80; % in-plane shear strength
22
23 % Layup and geometry
24 tp = 0.5; % ply thickness (mm)
25 TH = 67.8; % variable angle (deg)
26 AL = 30.0; % fixed angle (deg)
27 pang = [TH, -TH, AL, -AL, -AL, AL, -TH, TH]; % 8-ply symmetric layup
28 nPly = numel(pang);
29 H_lam = nPly * tp; % total thickness = 4 mm
30
31 % Applied laminate in-plane stresses (MPa)
32 Sxx_app = 2099;
33 Syy_app = 620;
34 Sxy_app = 507;
35
36 %%
37 % 2. REDUCED STIFFNESS Q (material coords, MPa)
38 %%
39 den = 1 - nu12*nu21;
40 Q11 = E1/den; Q22 = E2/den;
41 Q12 = nu12*E2/den; Q66 = G12;
42 Q_0 = [Q11 Q12 0; Q12 Q22 0; 0 0 Q66];
43
44 fprintf('===== REDUCED STIFFNESS Q (MPa) =====\n');
45 fprintf('  Q11=%9.2f  Q22=%9.2f  Q12=%9.2f  Q66=%9.2f\n\n', Q11, Q22, Q12, Q66);
46
47 %%
48 % 3. [A] MATRIX (N/mm) and its inverse
49 %%
50 A_mat = zeros(3, 3);
51 Qb = cell(1, nPly);
52 for k = 1:nPly
53     Qb{k} = fn_Qbar(pang(k), Q_0);
54     A_mat = A_mat + Qb{k} * tp;
55 end
56 A_inv = inv(A_mat);
57
58 fprintf('===== [A] MATRIX (N/mm) =====\n');
59 disp(A_mat);
60
61 %%

```

MATLAB Code (P1-P2)

```

184 % FIGURE 2: Bar chart - MS per ply (unique 4 plies)
185 %%
186 f2 = figure('Name','MS per Ply','Color','w','Position',[60 60 880 500]);
187 bar_data = [MSf_all(1:4); MSm_all(1:4)];
188 bh = bar(bar_data,'grouped');
189 bh(1).FaceColor = [0.08 0.18 0.44]; % deep navy
190 bh(2).FaceColor = [0.87 0.21 0.17]; % crimson
191 hold on;
192 yline(0,'k--','LineWidth',1.3);
193 yline(MS_FPF,'--','Color',[0.95 0.62 0.08],'LineWidth',2.0, ...
194     'Label',sprintf('MS_FPF = %.4f',MS_FPF),'LabelHorizontalAlignment','left');
195 set(gca,'XTickLabel',{'+\Theta(+67.8\circ)','-\Theta(-67.8\circ)', ...
196     '+\alpha(+30\circ)','-\alpha(-30\circ)'}, 'FontSize',11);
197 legend({'MS - Fiber Mode','MS - Matrix Mode'},'Location','best');
198 xlabel('Ply','FontSize',12);
199 ylabel('Margin of Safety','FontSize',12);
200 title(sprintf('MS per Ply (Hashin) \\\sigma_{xx}=%d, \\\sigma_{yy}=%d, \\\tau_{xy}=%d MPa', ...
201     Sxx_app, Syy_app, Sxy_app), 'FontSize',12,'FontWeight','bold');
202 grid on;
203 saveas(f2,'chart_ms_bars_matlab.png');
204
205 %%
206 % FIGURE 3: Failure Envelopes (Extra Credit)
207 % <sxx> vs <syy> at (a) tau_xy=0 and (b) tau_xy=507 MPa
208 %%
209 sxx_env = linspace(0.5, 350, 400);
210 tau_cases = [0, Sxy_app];
211
212 ply_clr = {'b','c','r',[0.85 0.70 0.00]};
213 ply_fe_lbl = {'+\Theta(+67.8\circ)','-\Theta(-67.8\circ)', ...
214     '+\alpha(+30\circ)','-\alpha(-30\circ)'};
215 case_ttl = {'Case (a): \tau_{xy} = 0 MPa', ...
216     sprintf('Case (b): \tau_{xy} = %.0f MPa [shear-governed]', Sxy_app)};
217
218 f3 = figure('Name','Failure Envelopes','Color','w','Position',[60 60 1360 580]);
219
220 for ci = 1:2
221     tau_fix = tau_cases(ci);
222     subplot(1,2,ci);
223     hold on;
224
225     % Per-ply envelopes (plies 1-4)
226     for pi = 1:4
227         syy_ply = arrayfun(@(sx) ...
228             fn_bisect_syy(sx, tau_fix, pi, pang, Qb, A_inv, H_lam, ...
229                 F1T, F1C, F2T, F2C, F12), sxx_env);
230         plot(sxx_env, syy_ply, '--', 'Color', ply_clr(pi), 'LineWidth', 1.8, ...
231             'DisplayName', ply_fe_lbl(pi));
232     end
233
234     % FPF envelope
235     syy_FPF = arrayfun(@(sx) ...
236         fn_bisect_syy(sx, tau_fix, 0, pang, Qb, A_inv, H_lam, ...
237             F1T, F1C, F2T, F2C, F12), sxx_env);
238     plot(sxx_env, syy_FPF, 'k-', 'LineWidth', 2.8, 'DisplayName','FPF (worst ply)');
239     patch([sxx_env, flipr(sxx_env)], [syy_FPF, zeros(1,numel(syy_FPF))], ...
240         'g', 'FaceAlpha', 0.15, 'EdgeColor', 'none', 'DisplayName','Safe region');
241
242     if ci == 2
243         % — FIX: applied load is far outside envelope; annotate instead —
244         text(0.50, 0.80, ...

```

```

123 fprintf(' <Rxx> = %10.3f MPa\n', Rxx_lam);
124 fprintf(' <Ryy> = %10.3f MPa\n', Ryy_lam);
125 fprintf(' <Txy> = %10.3f MPa\n', Txy_lam);
126
127 %%
128 % 7. MS_o vs THETA SWEEP (Alpha = 30 deg fixed)
129 %%
130 th_vec = linspace(5, 85, 161);
131 MS_o_vec = zeros(size(th_vec));
132
133 for ii = 1:numel(th_vec)
134     th_ii = th_vec(ii);
135     angs_ii = [th_ii,-th_ii,AL,-AL,-AL,AL,-th_ii,th_ii];
136     A_ii = zeros(3,3);
137     Qb_ii = cell(1,8);
138     for k = 1:8
139         Qb_ii{k} = fn_Qbar(angs_ii(k), Q_0);
140         A_ii = A_ii + Qb_ii{k} * tp;
141     end
142     A1_ii = inv(A_ii);
143     MS_o_vec(ii) = fn_MS_FPF(Sxx_app, Syy_app, Sxy_app, ...
144         angs_ii, Qb_ii, A1_ii, H_lam, ...
145         F1T, F1C, F2T, F2C, F12);
146
147 end
148 [MS_opt, idx_opt] = max(MS_o_vec);
149 th_opt = th_vec(idx_opt);
150 idx_given = find(abs(th_vec - TH) < 0.5, 1);
151 MS_given = MS_o_vec(idx_given);
152
153 fprintf('\n==== THETA OPTIMIZATION =====\n');
154 fprintf(' Given Theta = %.1f deg --> MS_o = %.4f\n', TH, MS_given);
155 fprintf(' Optimal Theta = %.1f deg --> MS_o,max = %.4f\n', th_opt, MS_opt);
156
157 %%
158 % FIGURE 1: MS_o = f(Theta)
159 %%
160 f1 = figure('Name','MS_o vs Theta','Color','w','Position',[60 60 960 530]);
161 plot(th_vec, MS_o_vec, 'b-', 'LineWidth', 2.2);
162 hold on;
163 yline(0,'k--','LineWidth',1.3,'Label','MS = 0 (failure threshold)');
164 plot(th_opt, MS_opt, 'ro','MarkerSize',11,'MarkerFaceColor','r');
165 plot(TH, MS_given,'bs','MarkerSize',11,'MarkerFaceColor','b');
166 text(th_opt + 1.5, MS_opt + 0.015, ...
167     sprintf('\Theta_{opt} = %.1f\circ\nMS_{o,max} = %.4f', th_opt, MS_opt), ...
168     'Color','r','FontSize',10,'FontWeight','bold');
169 text(TH - 14, MS_given + 0.015, ...
170     sprintf('\Theta_{given} = %.1f\circ\nMS_o = %.4f', TH, MS_given), ...
171     'Color','b','FontSize',10,'FontWeight','bold');
172 xlabel('Ply Angle \Theta (degrees)','FontSize',12);
173 ylabel('Margin of Safety MS_o','FontSize',12);
174 title(['+\Theta/-\Theta+30\circ/-30\circ_s SM Carbon | Hashin FPF | MS_o(\Theta)', ...
175     'FontSize',13,'FontWeight','bold']);
176 legend({'MS_o(\Theta)', 'Failure threshold', 'Optimal \Theta', 'Given \Theta'}, ...
177     'Location','best');
178 grid on;
179 xlim([0 90]);
180 set(gca,'FontSize',11);
181 saveas(f1,'chart_ms_theta_matlab.png');
182
183 %%

```

MATLAB Code (P3-P4)

```

306         n^2,    m^2,    ~2*m*n ;
307         ~m*n,   m*n,    m^2-n^2 ];
308     end
309
310     function [MSf, MSm, mode_f, mode_m] = fn_hashin(s1, s2, t12, ...
311                                                     FIT, F1C, F2T, F2C, F12)
312 %FN_HASHIN Hashin (1980) failure criterion - fiber and matrix margins
313
314 % --- Fiber mode ---
315 if s1 >= 0
316     FI_f = (s1/F1T)^2 + (t12/F12)^2;
317     mode_f = 'FT';
318 else
319     FI_f = (s1/F1C)^2;
320     mode_f = 'FC';
321 end
322 if FI_f <= 0
323     MSf = 1e9;
324 else
325     MSf = 1/sqrt(FI_f) - 1;
326 end
327
328 % --- Matrix mode ---
329 if s2 >= 0
330     FI_m = (s2/F2T)^2 + (t12/F12)^2;
331     mode_m = 'MT';
332     if FI_m <= 0
333         MSm = 1e9;
334     else
335         MSm = 1/sqrt(FI_m) - 1;
336     end
337 else
338     Ac = (s2/(2*F12))^2 + (t12/F12)^2;
339     Bc = ((F2C/(2*F12))^2 - 1) * (s2/F2C);
340     disc = Bc^2 + 4*Ac;
341     if disc < 0 || Ac == 0
342         MSm = 1e9;
343     else
344         lam1 = (-Bc + sqrt(disc)) / (2*Ac);
345         lam2 = (-Bc - sqrt(disc)) / (2*Ac);
346         lams = [lam1, lam2];
347         lams = lams(lams > 0);
348         if isempty(lams)
349             MSm = 1e9;
350         else
351             MSm = min(lams) - 1;
352         end
353     end
354     mode_m = 'MC';
355 end
356 end
357
358 function MS = fn_MS_FPF(sx, sy, sxy, ang, Qb_all, Ainv, H, ...
359                         FIT, F1C, F2T, F2C, F12)
360 %FN_MS_FPF FPF margin for a given in-plane stress state
361 N_v = [sx; sy; sxy] * H;
362 eps = Ainv * N_v;
363 MS = 1e9;
364 for k = 1:numel(ang)
365     sg_k = Qb_all{k} * eps;
366     sm_k = fn_Tstress(ang(k)) * sg_k;

```

```

245     sprintf(['Applied load:\n' ...
246             '\sigma_{xx} = %d MPa, \sigma_{yy} = %d MPa\n\n' ...
247             '\tau_{xy} = %d MPa >> F_{12} = %d MPa\n' ...
248             'Global shear alone exceeds ply shear strength.\n' ...
249             'Applied point lies FAR outside safe region.\n' ...
250             'Laminate has already failed under this loading.'], ...
251         Sxx_app, Syy_app, Sxy_app, F12), ...
252     'Units', 'normalized', ...
253     'HorizontalAlignment', 'center', ...
254     'VerticalAlignment', 'middle', ...
255     'FontSize', 10, 'Color', [0.75 0 0], 'FontWeight', 'bold', ...
256     'BackgroundColor', [1 0.93 0.93], ...
257     'EdgeColor', [0.75 0 0], 'LineWidth', 1.2);
258
259 end
260 xlabel('<\sigma_{xx}> (MPa)', 'FontSize', 11);
261 ylabel('<\sigma_{yy}> (MPa)', 'FontSize', 11);
262 title(case_ttl{ci}, 'FontSize', 12, 'FontWeight', 'bold');
263 legend('Location', 'northeast', 'FontSize', 8);
264 grid on;
265 xlim([0, max(sxx_env)]);
266 ymax_cur = max([syy_FPF, 50]);
267 ylim([0, ymax_cur * 1.15]);
268 set(gca, 'FontSize', 10);
269 end
270
271 sgtitle(sprintf(['+\Theta/-\Theta/+30\circ/-30\circ]_s SM Carbon - ' ...
272               'FPF Failure Envelopes (\Theta = %.1f\circ)', TH), ...
273         'FontSize', 13, 'FontWeight', 'bold');
274 saveas(F3, 'chart_failure_envelopes_matlab.png');
275
276 fprintf('\n==== ALL DONE - figures saved ==== \n');
277 fprintf(' chart_ms_theta_matlab.png\n');
278 fprintf(' chart_msBars_matlab.png\n');
279 fprintf(' chart_failure_envelopes_matlab.png\n');
280
281 %% =====
282 % LOCAL FUNCTIONS
283 %% =====
284
285 function Qb = fn_Qbar(ang_deg, Q)
286 %FN_QBAR Transformed reduced stiffness in global x-y frame
287 m = cosd(ang_deg);
288 n = sind(ang_deg);
289 Qb = zeros(3,3);
290 Qb(1,1) = Q(1,1)*m^4 + 2*(Q(1,2)+2*Q(3,3))*m^2*n^2 + Q(2,2)*n^4;
291 Qb(2,2) = Q(1,1)*n^4 + 2*(Q(1,2)+2*Q(3,3))*m^2*n^2 + Q(2,2)*m^4;
292 Qb(1,2) = (Q(1,1)+Q(2,2)-4*Q(3,3))*m^2*n^2 + Q(1,2)*(m^4+n^4);
293 Qb(2,1) = Qb(1,2);
294 Qb(1,3) = (Q(1,1)-Q(1,2)-2*Q(3,3))*m^3*n - (Q(2,2)-Q(1,2)-2*Q(3,3))*m*n^3;
295 Qb(3,1) = Qb(1,3);
296 Qb(2,3) = (Q(1,1)-Q(1,2)-2*Q(3,3))*m*n^3 - (Q(2,2)-Q(1,2)-2*Q(3,3))*m^3*n;
297 Qb(3,2) = Qb(2,3);
298 Qb(3,3) = (Q(1,1)+Q(2,2)-2*Q(1,2)-2*Q(3,3))*m^2*n^2 + Q(3,3)*(m^4+n^4);
299 end
300
301 function T = fn_Tstress(ang_deg)
302 %FN_TSTRESS Stress transformation matrix (Voigt notation)
303 m = cosd(ang_deg);
304 n = sind(ang_deg);
305 T = [ m^2,    n^2,    2*m*n ;

```

MATLAB Code (P5-P6)

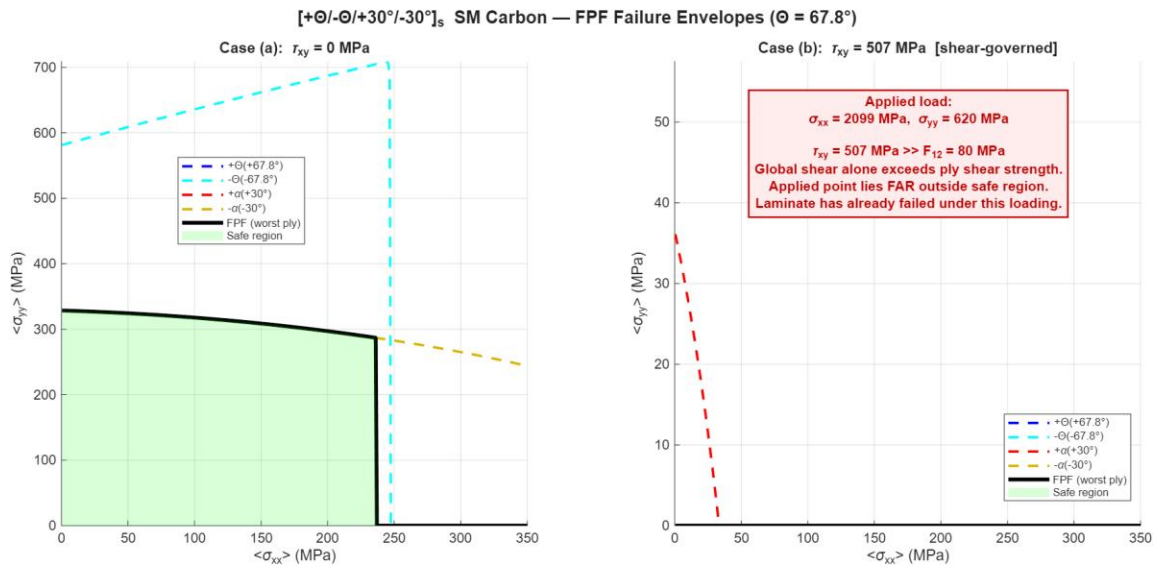
```

367         [MSf_k, MSm_k, ~, ~] = fn_hashin(sm_k(1), sm_k(2), sm_k(3), ...
368                                         F1T, F1C, F2T, F2C, F12);
369         MS = min(MS, min(MSf_k, MSm_k));
370     end
371 end
372
373 function syy_cr = fn_bisect_syy(sxx, sxy_fix, ply_id, angs, Qb_all, Ainv, H, ...
374                               F1T, F1C, F2T, F2C, F12)
375 %FN_BISECT_SYY Binary search: find syy where MS = 0
376     lo = 0;
377     hi = 8000;
378     for it = 1:60
379         mid = (lo + hi) / 2;
380         if ply_id == 0
381             ms = fn_MS_FPF(sxx, mid, sxy_fix, angs, Qb_all, Ainv, H, ...
382                           F1T, F1C, F2T, F2C, F12);
383         else
384             N_t = [sxx; mid; sxy_fix] * H;
385             ep_t = Ainv * N_t;
386             sg_t = Qb_all{ply_id} * ep_t;
387             sm_t = fn_Tstress(angs(ply_id)) * sg_t;
388             [MSf_t, MSm_t, ~, ~] = fn_hashin(sm_t(1), sm_t(2), sm_t(3), ...
389                                             F1T, F1C, F2T, F2C, F12);
390             ms = min(MSf_t, MSm_t);
391         end
392         if ms > 0
393             lo = mid;
394         else
395             hi = mid;
396         end
397     end
398     syy_cr = lo;
399 end

```

MATLAB Code (P7)

Failure Envelopes (Extra Credit)



Case (a): $\tau_{xy} = 0$ MPa

- Safe region (green) is bounded by ≈ 330 MPa in σ_{yy} and ≈ 245 MPa in σ_{xx} at FPF.
- The $-\Theta$ ply (cyan dashed) governs the upper σ_{yy} boundary.
- FPF envelope (solid black) is the most conservative of all individual ply envelopes.

Case (b): $\tau_{xy} = 507$ MPa (applied shear load)

- $\tau_{xy} = 507$ MPa $\gg T_{12} = 80$ MPa (ply shear strength).
- Global shear stress alone exceeds ply shear capacity.
- Safe region is negligible; the envelope collapses near the origin.
- Applied load ($\sigma_{xx} = 2099$ MPa, $\sigma_{yy} = 620$ MPa) lies far outside any feasible safe region.
- This result is consistent with $MS_0 = -0.9065$ and confirms shear-dominated failure.

Conclusions

Laminate Has Failed

All plies show negative margins. $MS_0 = -0.9065$ (FPF). Critical ply: -67.8° (matrix tension mode).

Shear is the Governing Load

Applied $\tau_{xy} = 507$ MPa is $6.34\times$ the ply shear strength (80 MPa). Failure envelope for $\tau_{xy} = 507$ MPa is nearly empty.

Angle Optimization

$\Theta_{\text{opt}} = 33.0^\circ$ gives $MS_{0,\text{max}} = -0.7078$, a 22% improvement over $\Theta = 67.8^\circ$ ($MS_0 = -0.9062$). The laminate still fails at all Θ under this load magnitude.

Design Recommendation

Redesign is required:

Reduce applied τ_{xy} through load path changes.

Increase laminate thickness or number of plies.

Consider a material with higher shear strength (though SM Carbon is already among the highest).

If retaining this layup family, adopt $\Theta_{\text{opt}} = 33^\circ$
